EBM TRANSFORMER COMPOSITIONAL GRASPING: A MUJOCO NEGATIVE RESULT

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ABSTRACT

This paper records a negative result for an energy-based Transformer mechanism for compositional grasping. The hypothesis was that grasp selection should compose separate object, contact, task, collision, and feasibility energies instead of collapsing a scene into one policy token. We rebuilt the idea as a MuJoCo parallel-jaw grasping benchmark with rollout-labeled training data, learned rankers, analytic force-closure and CEM baselines, ablations, stress sweeps, and seed-level uncertainty. The mechanism fails the submission gate. On the combined object/contact/task/clutter shift, the proposed EBM/Transformer ranker reaches 0.080 success, while force-closure and MLP energy baselines reach 0.140. Pairwise seed comparisons show no reliable advantage over strong non-oracle baselines. We therefore mark this project **KILL/ARCHIVE** for ICLR main rather than submit an unsupported compositional-grasping claim.

1 QUESTION

Transformer-based grasping, open-vocabulary affordance localization, and task-aware grasp planning are crowded areas (FViT-Grasp; OVAL-Grasp; HMT-Grasp). The original research bet here was specific: a grasp should not be represented by a single policy token. Instead, grasp ranking should compose object geometry, contact stability, task constraints, clutter/collision risk, and gripper feasibility as separate energy terms. That mechanism is appealing because grasp success often fails through one violated factor. The empirical question is whether explicit energy composition actually improves closed-loop grasping under held-out object and task compositions.

2 BENCHMARK

We implemented a compact MuJoCo parallel-jaw grasping benchmark. Candidate grasps are parameterized by planar center, approach angle, jaw width, closing force, and lift tilt. Each candidate is executed by closing two position-controlled capsule fingers around a free object and lifting it. The benchmark records lift success, slip, drop, collision with clutter, unsafe force, energy, and task satisfaction.

The seven main splits are seen simple objects, unseen dimensions, unseen shape family, slippery contact, clutter collision, task constraint shift, and combined composition shift. Objects include cuboids, cylinders, bars, and asymmetric L/T composites. Tasks include lift-only, avoid a forbidden face, grasp a long-side handle, and clutter-safe extraction.

We compare nine methods. `random_grasp` is a lower bound. `antipodal_geometry` uses geometric width and centering. `force_closure_score` adds friction and squeeze terms. `cem_grasp_search` uses a stronger analytic candidate score. `mlp_energy_model` is a trained scalar MLP over grasp/object/task features. `transformer_policy_ranker` is a small context-aware Transformer ranker trained on MuJoCo rollout labels. `ensemble_uncertainty_ranker` penalizes learned disagreement. `ebm_transformer_compositional` is the proposed method, composing learned context with object/contact/task/collision/feasibility energies. `oracle_mujoco_grid` evaluates six candidate grasps in MuJoCo and is a non-submission upper bound.

Table 1: Combined composition-shift results. The proposed method does not clear analytic or learned baselines.

Method	Succ.	95% CI	Slip	Drop	Collision	Energy
`random_grasp`	0.020	0.039	0.114	0.800	0.940	1.749
`antipodal_geometry`	0.080	0.076	0.094	0.560	0.840	3.188
`force_closure_score`	0.140	0.097	0.086	0.360	0.780	3.545
`cem_grasp_search`	0.100	0.084	0.081	0.360	0.820	3.960
`mlp_energy_model`	0.140	0.097	0.075	0.400	0.840	3.625
`transformer_policy_ranker`	0.060	0.067	0.076	0.420	0.900	2.737
`ensemble_uncertainty_ranker`	0.100	0.084	0.083	0.440	0.820	3.564
`ebm_transformer_compositional`	0.080	0.076	0.085	0.420	0.820	3.574
`oracle_mujoco_grid`	0.200	0.112	0.050	0.120	0.740	4.068

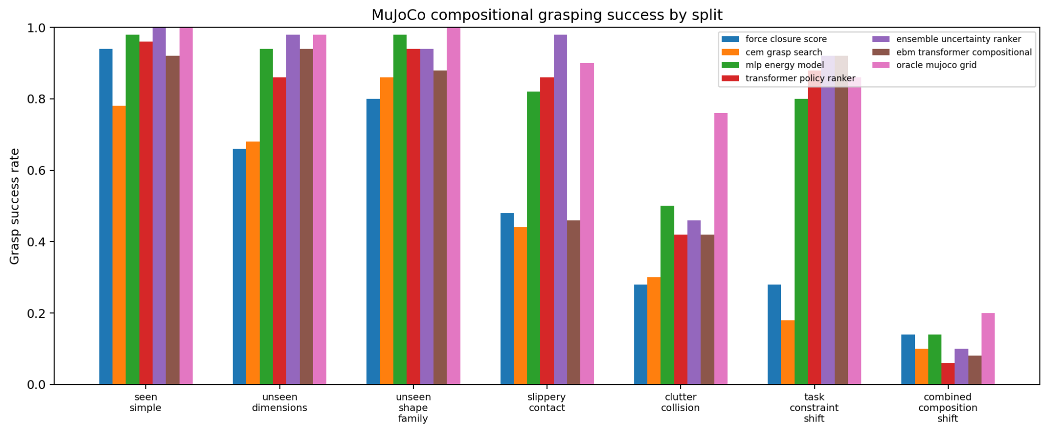

Figure 1: MuJoCo grasp success across object, contact, task, and clutter shifts. The EBM/Transformer ranker has no broad advantage.

3 Training and Evaluation

The training set contains 720 MuJoCo candidate rollouts with a 0.394 success rate. The main evaluation contains 3,150 MuJoCo rollouts: 5 seeds, 10 episodes per seed, 7 splits, and 9 methods. The ablation suite adds 400 rollouts and the stress sweep adds 1,200 rollouts. All reported success rates are measured by executed MuJoCo grasps, not by the learned model scores.

Table 1 is the central gate. On the combined composition shift, the proposed method reaches 0.080 success. Force closure and MLP energy both reach 0.140, while the oracle grid reaches 0.200 after evaluating six MuJoCo candidates. The oracle result shows that better selection is possible, but the proposed compositional ranker does not recover it. Pairwise seed comparisons give no reliable advantage over antipodal geometry, force closure, CEM, MLP energy, transformer ranking, or ensemble uncertainty.

4 Ablations

The ablations are mixed. The full energy composition is better than the monolithic scalar ablation in this run, which means the energy terms are not meaningless. However, this internal signal is not enough for ICLR-main readiness. The full mechanism must beat external analytic and learned baselines on the same held-out composition split, and it does not.

Table 2: Combined-shift ablations. The full method is better than several removals, but not enough to beat external baselines.

Ablation	Success	95% CI	Collision
full_ebm_transformer_compositional	0.180	0.108	0.740
no_transformer_context	0.100	0.084	0.840
no_collision_energy	0.080	0.076	0.900
no_contact_energy	0.080	0.076	0.840
no_object_energy	0.080	0.076	0.820
no_task_energy	0.080	0.076	0.840
no_feasibility_energy	0.060	0.067	0.880
monolithic_scalar_energy_only	0.040	0.055	0.920

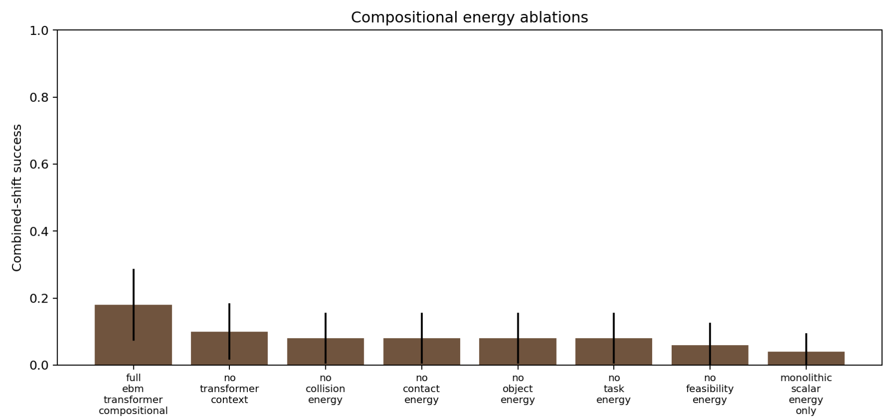

Figure 2: Ablations on the combined shift. Some energy terms matter internally, but the full method still fails the external baseline gate.

5 Stress and Safety

The stress sweep and safety plots explain the failure mode. The hardest compositions are cluttered, slippery, and geometrically asymmetric. The proposed ranker often chooses geometrically plausible grasps that still collide with clutter or drop during lift. Explicit energy terms help describe these factors, but the learned scoring and candidate search are not strong enough to produce a decisive success-safety frontier.

6 Limitations

This is a MuJoCo simulator study rather than hardware or a public grasp benchmark. The visual input is abstracted into object and candidate features, not raw RGB-D perception. The Transformer ranker is deliberately small to keep the run reproducible on CPU. The related-work audit is still based on the local hostile pool rather than a full manual paper-by-paper synthesis. Those limitations would support a strong-revise outcome if the mechanism had clearly beaten baselines. It did not, so the honest terminal action is archive.

7 Decision

The submission gate required the EBM/Transformer composition to beat force closure, CEM, MLP energy, transformer ranking, ensemble uncertainty, and monolithic-energy ablations under held-out composition. It failed that gate. The correct decision is **KILL/ARCHIVE**. Future work would need richer perception, tactile/material sensing, larger public grasp benchmarks, and a stronger candidate-search procedure before reviving the idea.

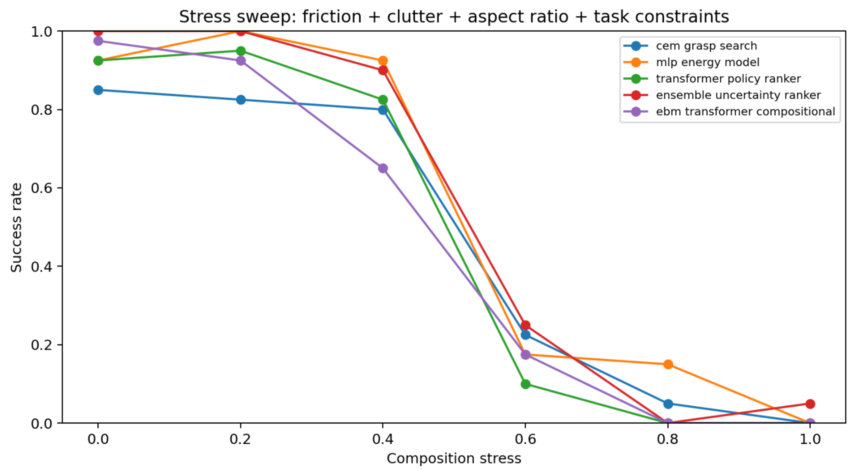

Figure 3: Stress sweep over friction, clutter, aspect ratio, and task constraint strength. The proposed ranker is not a robust frontier.

Figure 4: Collision plus unsafe-force failures. The hardest splits are dominated by clutter collisions, not solved by the proposed energy composition.

References

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